

Bahari Hai Anti-Poaching

Workflow User Guide

Marine Patrol Dashboard & Report

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This guide walks you through configuring and running the Bahari Hai Anti-Poaching Workflow, which transforms coastal and marine patrol data into a comprehensive anti-poaching and patrol effort report, interactive dashboard, and geospatial hotspot maps.

Overview

The **BH Anti-Poaching Workflow** processes patrol findings data from the Bahari Hai coastal and marine conservation area to produce a structured patrol report and an interactive dashboard. It analyses patrol effort by transport type (Boat, Foot, Vehicle), illegal activity patterns across villages and sectors, and temporal trends from year-level down to monthly resolution.

The workflow analyses:

- **Patrol Effort** — patrol counts, hours, and incident rates by year and transport type (Boat / Foot / Vehicle).
- **Illegal Activities** — Mangrove Logging, Poaching, Illegal Fishing, Damaging Coral, and Arrests.
- **Evidence Breakdowns** — detailed counts for mangrove logging, poaching gear, and illegal fishing evidence.
- **Village & Sector Summaries** — proportional effort, arrests by village, and top-5 village rankings.
- **Temporal Trends** — year, quarter, and monthly time-series charts.
- **Geospatial Hotspot Maps** — illegal events, patrol effort (weighted, standardised, ecograph), and five activity-specific maps.
- **Automated Word Report** (patrol_report.docx) — all visuals assembled into one document.
- **Interactive Dashboard** — filterable by the temporal grouper you configure (recommended: Year).

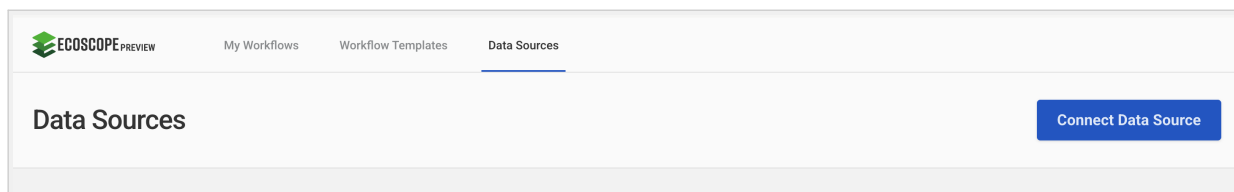
Prerequisites: Connect a Data Source

■ A data source **MUST** be configured in EcoScope Desktop before this workflow can run. The patrol CSV, marine layers, and report template are all fetched automatically — but the EcoScope platform requires an active data source connection to authenticate and execute any workflow. Set this up once and it applies to all workflows.

If you have already connected your EarthRanger site, skip to Step 1.

A Open the Data Sources tab

Click **Data Sources** in the top navigation bar, then click **Connect Data Source**.



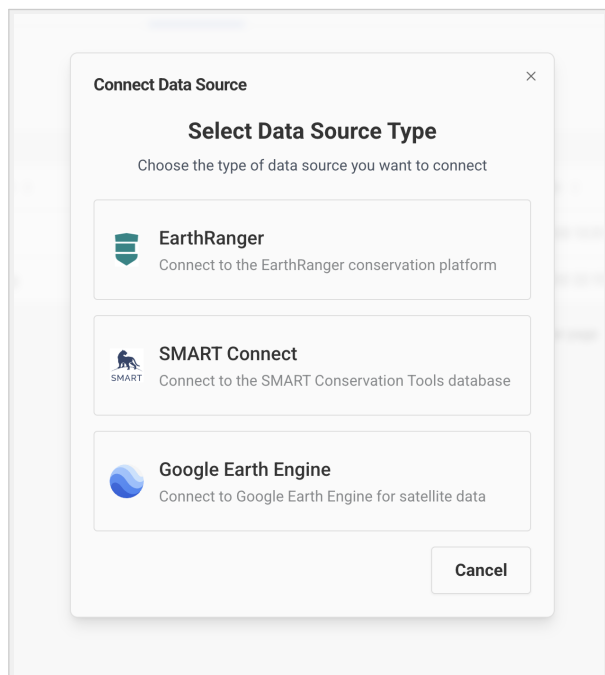
The Data Sources tab — click Connect Data Source.

B Select EarthRanger

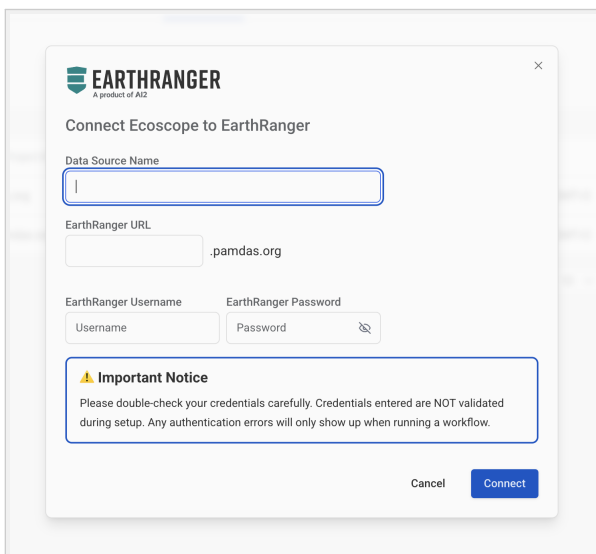
Click **EarthRanger** in the Select Data Source Type dialog.

C Enter your credentials and connect

Fill in Data Source Name, EarthRanger URL, Username, and Password, then click **Connect**.



Select EarthRanger as the data source type.



Enter credentials and click Connect.

Credentials are not validated during setup — authentication errors only appear when a workflow runs.

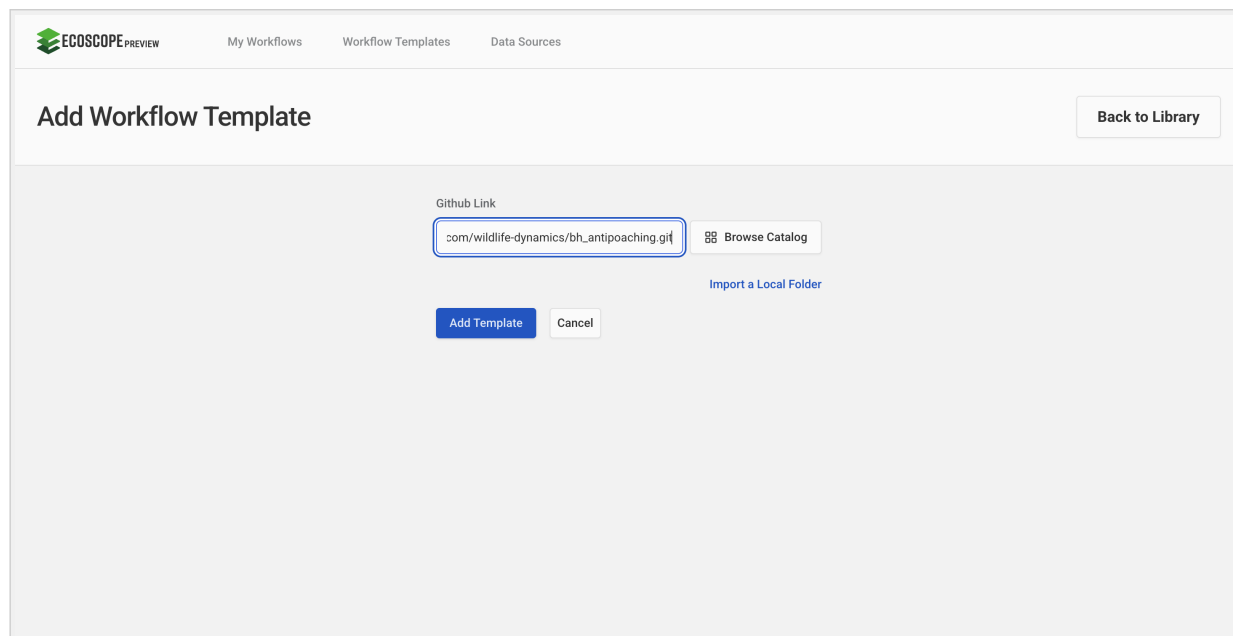
Step-by-Step Configuration

1 Add the Workflow Template

Click **Workflow Templates**, then click **Add Workflow Template**. Paste the GitHub URL into the **Github Link** field:

```
https://github.com/wildlife-dynamics/bh_antipoaching.git
```

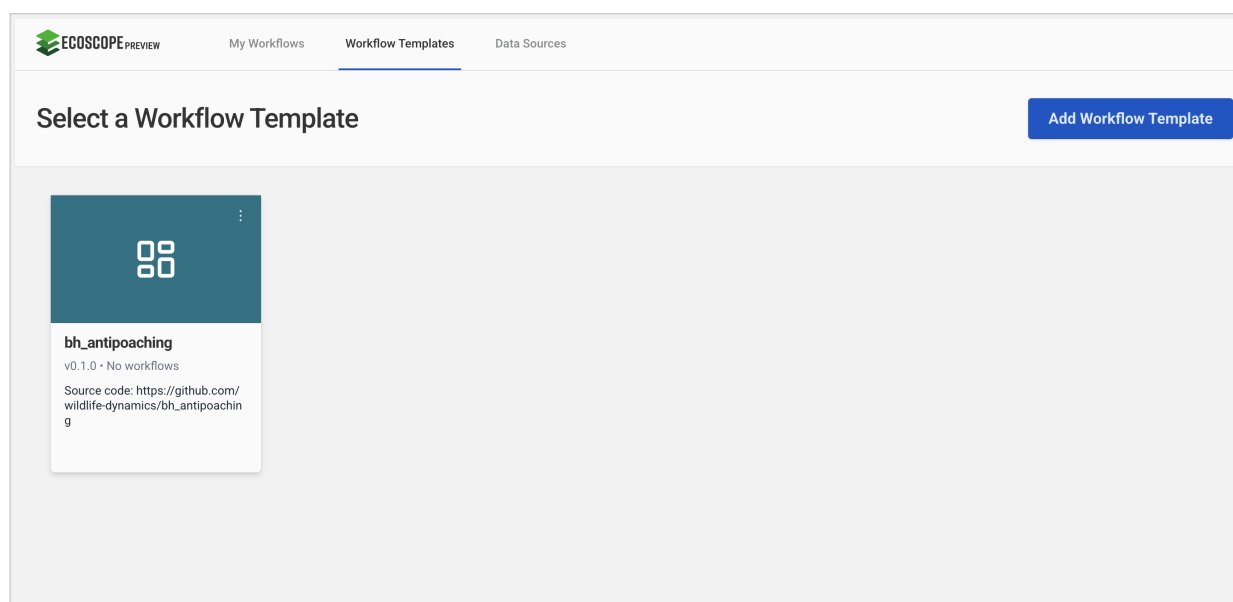
Click **Add Template**.



Pasting the bh_antipoaching GitHub URL into the Add Workflow Template page.

2 Select the Workflow

The **bh_antipoaching** card appears in the Workflow Templates list. Click it to open the configuration form.



The bh_antipoaching workflow card in the Workflow Templates list.

3 Set Workflow Details and Time Range

The configuration form has **three sections** shown in the left sidebar: *Set Workflow Details*, *Define analysis time range*, and *Define temporal groupers*.

Fill in the top two sections:

Field	Description
Workflow Name	e.g. Bahari Hai Anti-Poaching – Q2 2026
Workflow Description	Optional notes about the reporting period
Timezone	Africa/Nairobi (UTC+03:00)
Since	Start date of the analysis period
Until	End date of the analysis period

The screenshot shows a web interface for configuring a workflow. On the left is a sidebar with three menu items: 'Set Workflow Details', 'Define analysis time range', and 'Define temporal groupers'. The main content area is divided into two sections. The top section, 'Set Workflow Details', has a subtitle 'Add information that will help to differentiate this workflow from another.' and contains two text input fields: 'Workflow Name*' and 'Workflow Description'. The bottom section, 'Define analysis time range', has a subtitle 'Choose the period of time to analyze.' and contains a 'Timezone' dropdown menu (set to 'Africa/Nairobi (UTC+03:00)'), a 'Since' date field (set to '09/05/2026, 00:00' with the label 'The start time'), and an 'Until' date field (set to '08/06/2026, 23:59' with the label 'The end time').

The configuration form — Set Workflow Details and Define analysis time range sections.

4 Define Temporal Groupers — Group the Dashboard by Year

This is the step unique to this workflow. Scroll down to the **Define temporal groupers** section. This setting controls how the dashboard splits its widgets — without it, all years appear in a single combined view.

■ **Always add a temporal grouper. Leaving it blank merges all years together, making year-by-year comparison impossible in the dashboard.**

Follow these three sub-steps:

a Click Add

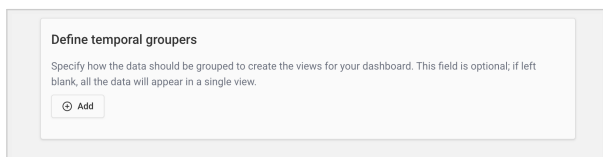
Click the + **Add** button. A new row appears with two dropdowns.

b Leave Group by as 'Category' (default)

The **Group by** dropdown defaults to **Category**. Do not change it.

c Set Category to 'Year'

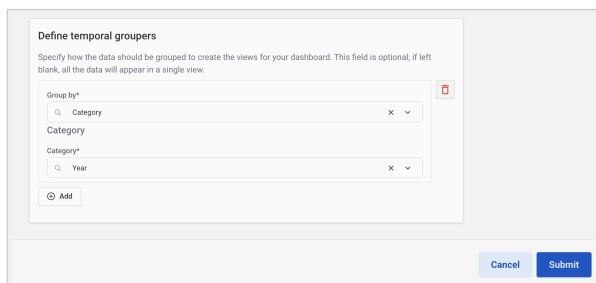
In the **Category** dropdown below, select **Year**. This groups all dashboard widgets by calendar year.



Define temporal groupers

Specify how the data should be grouped to create the views for your dashboard. This field is optional; if left blank, all the data will appear in a single view.

+ Add



Define temporal groupers

Specify how the data should be grouped to create the views for your dashboard. This field is optional; if left blank, all the data will appear in a single view.

Group by*
Category

Category*
Year

+ Add

Cancel Submit

The Define temporal groupers section — click Add to begin.

After clicking Add: leave Group by as Category, set Category to Year, then click Submit.

What Happens When You Submit

After clicking Submit the workflow runs automatically:

- Fetches the patrol findings CSV from GitHub.
- Fetches the marine layer shapefiles (ZIP) and report template (DOCX) from Google Drive.
- Cleans and parses the patrol data — normalises dates, extracts Year and Quarter fields.
- Splits data by the temporal grouper (Year) for year-by-year dashboard views.
- Generates all patrol summary tables and trend charts (patrols, hours, incidents per year).
- Generates village patrol summary table and patrol effort breakdown by transport type.
- Generates illegal activity charts — summary dashboard, donut chart, evidence breakdowns.
- Generates village and sector analytics — proportional effort, arrests, top-5 rankings, monthly trends.
- Renders all 9 geospatial hotspot and effort maps.
- Assembles the final Word report (patrol_report.docx) with all visuals embedded.
- Publishes the interactive dashboard, filterable by Year.

The workflow generates approximately 60+ outputs. Allow 5–10 minutes for a full run.

Outputs

All outputs are written to the directory set by ECOSCOPE_WORKFLOWS_RESULTS.

Patrol Summary Charts & Tables

Output	Description
patrol_summary.*	Multi-year patrol counts, hours, and incidents table
patrols_per_year.*	Line chart — number of patrols per year
patrol_hours_per_year.*	Line chart — patrol hours per year
incidents_per_year.*	Line chart — incidents per year
village_summary.*	Patrol summary by village
patrol_effort_by_transport.*	Stacked bar — patrol hours by transport type (Boat/Foot/Vehicle)
patrol_count_by_transport.*	Patrol count by transport type
patrol_hours_by_transport.*	Patrol hours by transport type

Illegal Activity Charts

Output	Description
illegal_activity_summary.*	Illegal activity summary dashboard
illegal_activity_donut.*	Donut chart of illegal activity categories
mangrove_evidence.*	Mangrove logging evidence breakdown
poaching_evidence.*	Poaching evidence breakdown
illegal_fishing_evidence.*	Illegal fishing gear evidence breakdown
illegal_events_by_sector.*	Stacked bar — illegal events by sector
arrests_by_village.*	Arrests by village chart
top_villages_category.*	Top 5 villages by activity category
top_villages_monthly.*	Top 5 villages monthly trend
village_monthly_timeseries.*	Village-level monthly time series

Geospatial Maps

Output	Description
illegal_events_map.*	All illegal events hotspot map
weighted_effort_map.*	Weighted patrol effort map
standardised_rate_map.*	Standardised illegal event rate map
ecograph_map.*	Ecograph patrol effort map
mangrove_logging_map.*	Mangrove logging hotspot map
poaching_map.*	Poaching hotspot map
illegal_fishing_map.*	Illegal fishing hotspot map
arrests_map.*	Arrests hotspot map
poached_turtles_map.*	Poached turtles hotspot map

Final Report

File	Description
patrol_report.docx	Complete Word report with all patrol, illegal activity, and map visuals embedded

For support, open an issue on the [wildlife-dynamics/bh_antipoaching](#) GitHub repository.